

Task 1 - Laying the Foundation for Analysis

Introduction

As a data scientist at Birhan Energies, the objective of this project is to analyze how major geopolitical, economic, and OPEC-related events impact Brent oil prices over the period 2012–2022. Task 1 focuses on defining a structured data analysis workflow, compiling a dataset of key events, identifying assumptions and limitations, and understanding the properties of the Brent oil price data and the Bayesian Change Point model. This interim report outlines the planned steps to achieve these goals, ensuring alignment with the business objective of providing actionable insights for investors, policymakers, and energy companies.

Data Analysis Workflow

The following steps outline the planned approach for analyzing Brent oil prices and their relationship with significant events:

Data Collection and Preparation:

Load the Brent oil price dataset (daily prices from May 20, 1987, to September 30, 2022) from the provided CSV file.

Convert the Date column to datetime format (from 'day-month-year', e.g., '20-May-87').

Filter data to focus on 2012–2022 to align with the project's scope.

Handle missing values (e.g., interpolate or flag for review) and ensure data consistency.

Compile a dataset of 10–15 major events (geopolitical, economic, OPEC policies) from 2012–2022, sourced from news archives, OPEC reports, and energy market databases (e.g., EIA, IEA).

Exploratory Data Analysis (EDA):

Visualize raw Brent oil prices to identify trends (e.g., 2014 price drop, 2020 COVID-19 crash) and periods of high volatility.

Compute log returns ($\log(\text{Price}_t) - \log(\text{Price}_{t-1})$) to transform non-stationary prices into a stationary series for modeling.

Conduct stationarity tests (Augmented Dickey-Fuller) to confirm properties of prices and log returns.

Plot log returns to observe volatility clustering, informing model selection.

Event Data Compilation:

Create a CSV file (events.csv) with columns: Event_Date, Event_Description, Event_Type.

Ensure events are relevant to oil price fluctuations (e.g., OPEC production decisions, geopolitical conflicts, economic shocks).

Validate event dates against historical records for accuracy.

Model Planning:

Plan to implement a Bayesian Change Point model using PyMC3 to detect structural breaks in the price series (e.g., shifts in mean price or volatility).

Define priors: a discrete uniform prior for the switch point (τ) and normal priors for mean prices before and after (μ_1 , μ_2).

Specify a normal likelihood to connect the model to observed prices.

Plan to use Markov Chain Monte Carlo (MCMC) sampling to estimate posterior distributions.

Insight Generation:

Align detected change points with compiled events to hypothesize relationships (e.g., 2020 COVID-19 demand shock causing a price drop).

Quantify impacts (e.g., percentage change in mean price at change points).

Address correlation vs. causation by noting that event alignment suggests association, not definitive causality.

Communication of Results:

Develop an interactive dashboard using Flask (backend) to serve data and React (frontend) with Chart.js for visualizations.

Include features like price charts, event annotations, and filters for date ranges or event types.

Prepare a concise report or Medium blog post summarizing findings, targeting investors (for risk management), policymakers (for energy security), and energy companies (for operational planning).

Host the dashboard online (e.g., Heroku) and share the report via email or publication platforms.

Assumptions and Limitations

Assumptions:

The Brent oil price dataset is complete with minimal missing data, allowing reliable analysis.

Selected events are representative of major influences on oil prices.

The Bayesian model's assumption of discrete change points is suitable for detecting abrupt price shifts.

Limitations:

Correlation vs. Causation: Change points aligned with events indicate correlation, but proving causation requires additional evidence (e.g., policy analysis).

Data Constraints: The analysis excludes external factors (e.g., GDP, exchange rates) due to dataset limitations, potentially missing broader influences.

Model Simplicity: A single change point model may overlook gradual changes or multiple simultaneous events.

Event Selection: The choice of 10–15 events may miss minor but relevant events, introducing selection bias.

Time Series Properties

The Brent oil price dataset will be analyzed for key properties:

Trend: Prices exhibit trends (e.g., decline in 2014–2016, recovery post-2020) driven by supply, demand, and external shocks.

Stationarity: Raw prices are expected to be non-stationary (ADF p-value > 0.05). Log returns will be computed to achieve stationarity (expected ADF p-value < 0.05), making them suitable for modeling.

Volatility: Log returns will reveal volatility clustering (e.g., high volatility during 2014 OPEC decisions, 2020 COVID-19 crash), guiding the choice of modeling volatility or mean shifts.

Impact on Modeling: Stationarity of log returns supports the use of a Bayesian Change Point model, as it ensures consistent statistical properties.

Change Point Model Overview

Purpose: To detect structural breaks in Brent oil prices (e.g., shifts in mean or volatility) caused by events like OPEC policy changes or geopolitical conflicts.

Mechanism: Using PyMC3, define a switch point (τ) with a discrete uniform prior and model mean prices (μ_1 , μ_2) before and after (τ). A normal likelihood connects the model to data, and MCMC sampling estimates posteriors.

Expected Outputs:

Change point dates (τ) with uncertainty (posterior distributions).

Parameter estimates (e.g., mean price before/after, standard deviation).

Probabilistic statements (e.g., “95% probability the mean price dropped after the change point”).

Limitations: Assumes abrupt changes, potentially missing gradual shifts, and requires sufficient data around change points for reliable detection.

Next Steps

Task 2: Implement the Bayesian Change Point model using PyMC3, interpret results, and associate change points with events.

Task 3: Develop the Flask-React dashboard to visualize price trends, change points, and event impacts.

Deliverables: Submit the final report, GitHub code, and dashboard screenshots by August 5, 2025.

This workflow ensures a robust foundation for analyzing Brent oil price fluctuations, aligning with the needs of Birhan Energies' stakeholders.